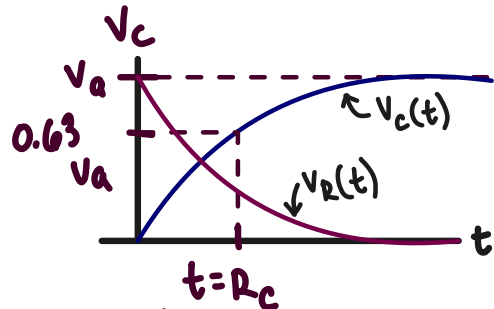
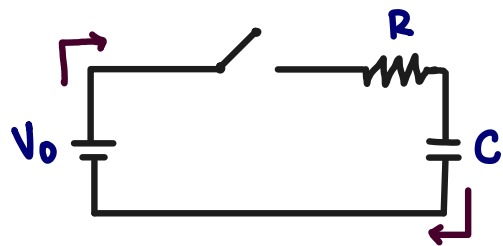


Charging a capacitor



$$V_C = V_0(1 - e^{-t/RC})$$

$$V_R = I(t)R = V_0 e^{-t/RC}, RC = \text{time constant}$$

Loop Rule:

$$V_0 - I(t)R - V_C(t) = 0$$

$$\text{recall, } I(t) = C \frac{dV_C}{dt}$$

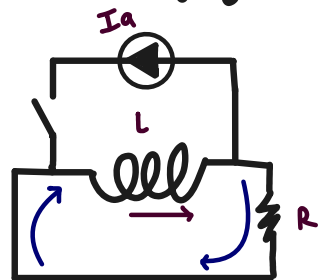
$$\text{so, } V_0 - RC \frac{dV_C}{dt} - V_C(t) = 0$$

$$V_C(0) = \frac{Q(0)}{C} = 0$$

1st order ODE

capacitor initially $Q(0)=0 \Rightarrow V_0 - R\left(\frac{A}{\tau}e^{-t/\tau} - A(I - e^{-t/\tau})\right) = 0$
 $\Rightarrow V_0 - R\left(\frac{A}{\tau}e^{-t/\tau} - A(I - e^{-t/\tau})\right) = 0$
 must be true @ all times
 as, $t \rightarrow \infty: e^{-t/\tau} \rightarrow 0$
 $V_0 - A = 0 \Rightarrow A = V_0$
 $\times t=0, e^{t/\tau} = 1$
 $V_0 - RC \frac{V_0}{\tau} 1 - V_0(I - 1) = 0$
 $\Rightarrow V_0 - RC \frac{V_0}{\tau} - V_0(I - 1) = 0$
 $\Rightarrow V_0 - RC \frac{V_0}{\tau} - V_0I + V_0 = 0$
 $\Rightarrow 2V_0 - RC \frac{V_0}{\tau} - V_0I = 0$
 $\Rightarrow I = \frac{2V_0}{RC} - \frac{V_0}{\tau}$
 * Guess a solⁿ *
 $V_C(t) = A + Be^{-t/\tau}$
 find A, B, τ
 $0 = V_C(0) = A + Be^0 = A + B$
 $\Rightarrow A = -B$
 so, $V_C(t) = A(I - e^{-t/\tau})$
 $\frac{dV_C}{dt} = -A(-\frac{1}{\tau})e^{-t/\tau} = \frac{A}{\tau}e^{-t/\tau}$

discharging an Inductor



Loop Rule: (switch open)
 $-V_L - V_R = 0$, recall: $V_L = L \frac{dI}{dt}$
 $V_R = IR$

$$\Rightarrow -L \frac{dI}{dt} - IR = 0$$

$$\frac{dI}{dt} = -\frac{R}{L}I \quad \text{1st order ODE}$$

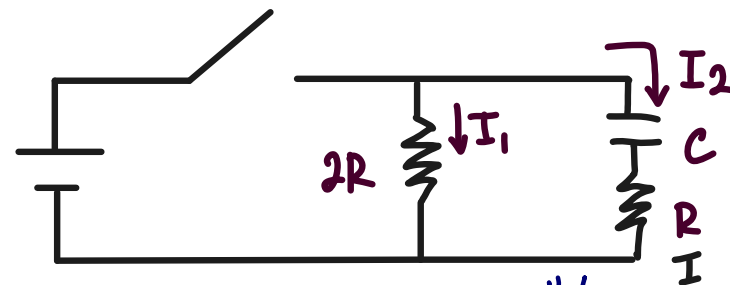
$$\frac{dI}{dt} = -\frac{R}{L}I$$

$$\int_{I(0)=I_0}^{I(t)} \frac{dI}{I} = \int_{t=0}^t -\frac{R}{L} dt = -\frac{R}{L} \int_{t=0}^t dt = -\frac{R}{L}t$$

$$\ln(I) \Big|_{I_0}^{I(t)} = \ln(I(t)) - \ln(I_0)$$

$$= \ln\left(\frac{I(t)}{I_0}\right) = -\frac{R}{L}t$$

$$\frac{I(t)}{I_0} = e^{-\frac{R}{L}t}, I(t) *$$



@ $t=0$, closed

$t=3RC$, opened back up

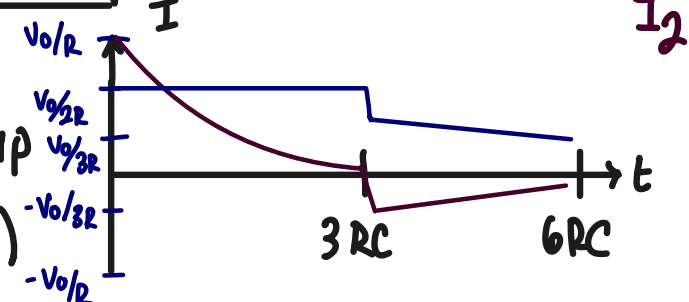
sketch $I_1(t)$ and $I_2(t)$

@ $t=0, V_C = 0$

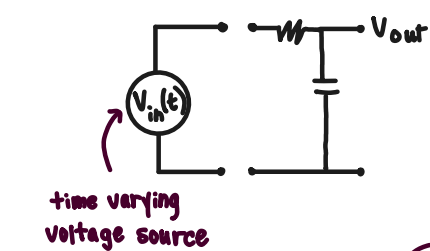
$$\Rightarrow \frac{V_0}{R} = I_1, I_1 = \frac{V_0}{2R}, I_2 = \frac{V_0}{R}$$

@ $t=3RC - \Delta t, V_C \approx V_0$

@ $t=3RC + \Delta t$ (after flip)



Integrators and Differentiators



Loop Rule: $V_{in}(t) - I(t)R - V_C(t) = 0$
 $V_C = \frac{1}{C} \int I dt$

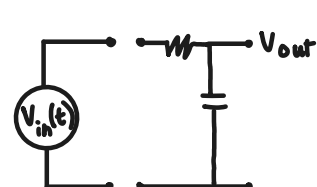
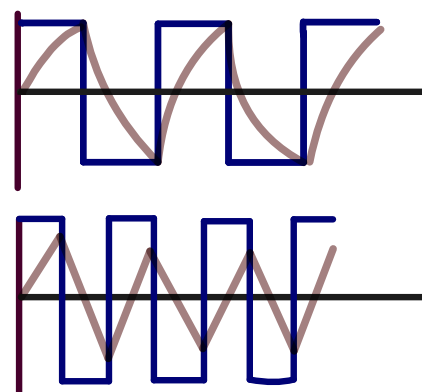
$$V_{in}(t) - RC \frac{dV_{out}}{dt} - V_{out}(t) = 0$$

• suppose $V_{in}(t)$ varies quickly
 over a time scale RC
 ex: V_{in} is periodic, (sine, square) w/ period $\ll RC$

$$V_{out} \propto \cos(\omega t), \frac{1}{\omega} \ll RC \Rightarrow \omega RC \gg 1$$

$$RC \frac{dV_{out}}{dt} \propto \omega RC \sin(\omega t)$$

and we can approximate loop rule



Loop Rule

$$V_{in} - V_C - V_{out} = 0$$

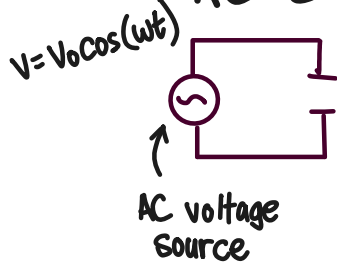
$$\leftarrow IR$$

* if $V_{in}(t)$ varies slowly

on timescale RC

$$V_{out} = \cos(\omega t) \quad \frac{dV_{out}}{dt} = \omega \sin(\omega t)$$

AC circuits and Complex Impedance



$$I(t) = C \frac{dV}{dt} = CV_0 \frac{d}{dt} \cos(\omega t)$$

$$= -CV_0 \omega \sin(\omega t)$$

$$= CV_0 \omega \cos\left(\omega t + \frac{\pi}{2}\right)$$

* more convenient to use complex #s

recall: $e^{i\phi} = \cos\phi + i\sin\phi$

$$\text{Re}\{e^{i\phi}\} = \cos\phi, \text{Im}\{e^{i\phi}\} = \sin\phi$$

↑ real part

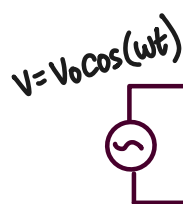
$$V(t) = \text{Re}\{V_0 e^{i\omega t}\} = V_0 \cos(\omega t)$$

$$\text{notation: } \tilde{V}(t) = V_0 e^{i\omega t}$$

$$\tilde{I}(t) = C \frac{d\tilde{V}}{dt} = CV_0 i\omega e^{i\omega t} = i\omega C \tilde{V}(t)$$

$$\tilde{V}(t) = Z_C \tilde{I}(t), Z_C = \frac{1}{i\omega C}$$

Z_C = complex impedance of a capacitor



$$V(t) = L \frac{dI}{dt}$$

$$\int \frac{V_0}{L} \cos(\omega t) dt \rightarrow \int \frac{dI}{dt} dt$$

$$\frac{V_0}{L} \frac{\sin(\omega t)}{\omega} = I(t)$$

$$\tilde{V}(t) = V_0 e^{i\omega t}$$

$$\tilde{V}(t) = L \frac{d\tilde{I}}{dt}$$

$$\int \frac{V_0}{L} e^{i\omega t} dt = \int \frac{d\tilde{I}}{dt}$$

Impedance

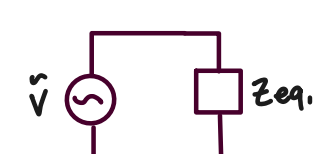
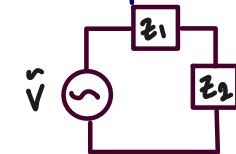
$\tilde{V} = \tilde{I} Z$, looks like complex version of Ohm's Law

$$Z_R = R, Z_C = \frac{1}{i\omega C}, Z_L = i\omega L$$

Rules for Equivalent Impedance

• Impedance in series add

• Impedance in || add like resistors



$$Z_{eq} = Z_1 + Z_2$$

Loop Rules

$$\tilde{V} - \tilde{I} Z_1 - \tilde{I} Z_2 = 0$$

$$\tilde{V} - \tilde{I} (Z_1 + Z_2) = 0$$